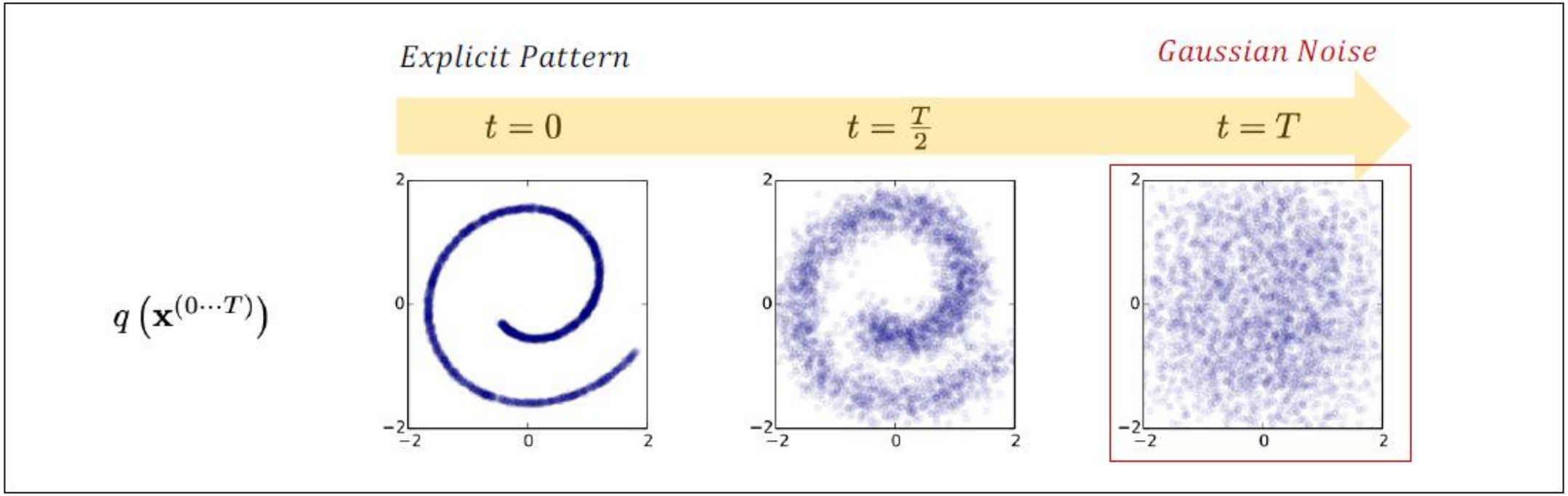
Diffusion 경량화 기법

권준상

목차

1. Diffusion
2. Distillation
3. Pruning
4. Quantization
5. TODO

Diffusion – Diffusion Process



https://velog.io/@ese2o/DDPM-Denoising-Diffusion-Probabilistic-Models

DATA -> Gaussian Noise 과정.
이 과정을 거꾸로 시행할 수 있으면 임의 생성된 Gaussian Noise -> DATA 가능하지 않을까?

Diffusion – DDPM 기본 구조

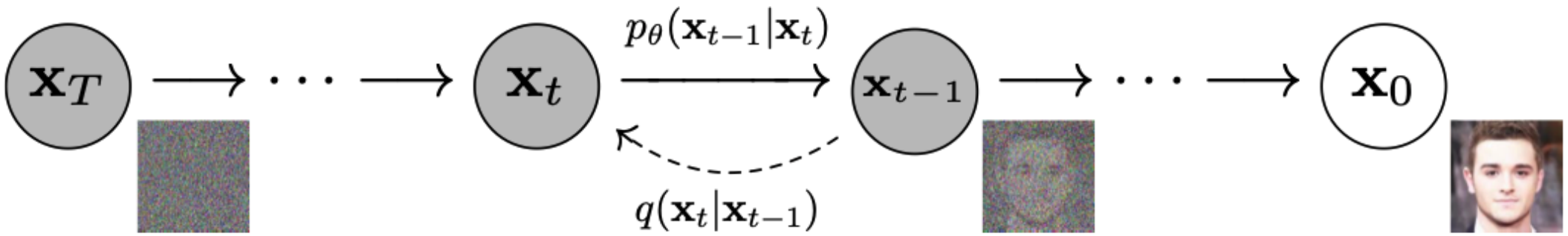


Figure 2: The directed graphical model considered in this work.

$q(x_t|x_{t-1})$: 정방향 확산 과정.

$p_\theta(x_{t-1}|x_t)$: $q(x_{t-1}|x_t)$를 근사(신경망 사용하여 파라미터 θ 학습 -> DDPM에선 Unet)

$p_\theta(x_{t-1}|x_t)$를 잘 예측(학습)한다면, pure gaussian noise로부터 원하는 이미지 예측 가능할 것

https://arxiv.org/pdf/2006.11239

Diffusion – DDPM($q(x_t|x_0)$)

using the notation $\alpha_t := 1 - \beta_t$ and $\bar{\alpha}_t := \prod_{s=1}^{t} \alpha_s$, we have

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$$

β_t : noise schedule, 각 timestep당 얼마만큼의 noise가 들어갈 것인가?

이로써 $q(x_t|x_0)$ 는 DATA($= x_0$)로부터 t번이 아닌 한번의 과정으로 어떤 timestep t로든 갈 수 있다.
+) t가 충분히 크다면 x_t는 mean 0, variance 1의 gaussian으로 수렴한다.

https://arxiv.org/pdf/2006.11239

Diffusion – DDPM Loss (1/4)

$$\mathbb{E}\left[-\log p_\theta(\mathbf{x}_0)\right] \leq \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t\geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right] =: L$$

Jenson's inequality

갖은 식변형 후..

Negative log likelihood of our prediction

https://arxiv.org/pdf/2006.11239

Diffusion – DDPM Loss (2/4)

$$\mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t\geq 1}\log\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right] =$$

$$\mathbb{E}_q\left[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)\parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1}\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{-\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0}\right]$$

Theta에 영향 x -> 상수이므로 제거

실제로 유의미

x_1 -> x_0 항인데...
영향이 적어서 실제 loss에서 무시
실제 논문 실험결과도 없어도 상관없다고 나옴

https://arxiv.org/pdf/2006.11239

Diffusion – DDPM Loss (3/4)

$$\mathbb{E}_q\Big[\underbrace{\cancel{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)\,\|\,p(\mathbf{x}_T))}}_{L_T} + \sum_{t>1}\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\,\|\,p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{\cancel{-\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}}_{L_0}\Big]$$

KL divergence 식 풀고 정리하면:

$$L_{t-1} = \mathbb{E}_q\left[\frac{1}{2\sigma_t^2}\|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t,\mathbf{x}_0) - \boldsymbol{\mu}_\theta(\mathbf{x}_t,t)\|^2\right] + C$$

Mean of $q(x_{t-1}|x_t,x_0)$

$$\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}\right)$$

μ_θ : $\tilde{\mu}_t$ 예측($p_\theta(x_{t-1}|x_t)$ 사용)

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t,t) = \tilde{\boldsymbol{\mu}}_t\left(\mathbf{x}_t, \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t))\right) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t,t)\right)$$

where $\boldsymbol{\epsilon}_\theta$ is a function approximator intended to predict $\boldsymbol{\epsilon}$ from $\mathbf{x}_t$.

우리가 최종적으로 가지고 싶은것
(얼마만큼의 noise가 껴있는가?)

https://arxiv.org/pdf/2006.11239

Diffusion – DDPM Loss (4/4)

$$L_{t-1} = \mathbb{E}_q\left[\frac{1}{2\sigma_t^2}\|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_\theta(\mathbf{x}_t, t)\|^2\right] + C$$

Noise image(step t)

$$L_{\text{simple}}(\theta) := \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t)\right\|^2\right]$$

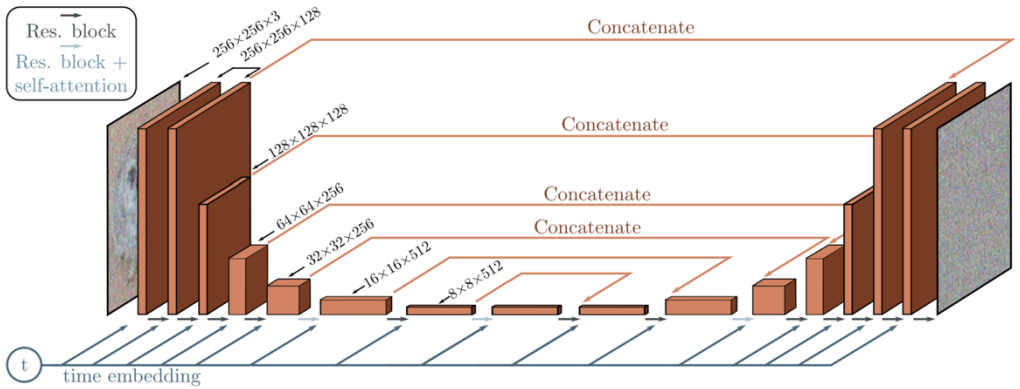
Timestep

이 값을 찾고 업데이트 하고 싶다..
-> Unet!

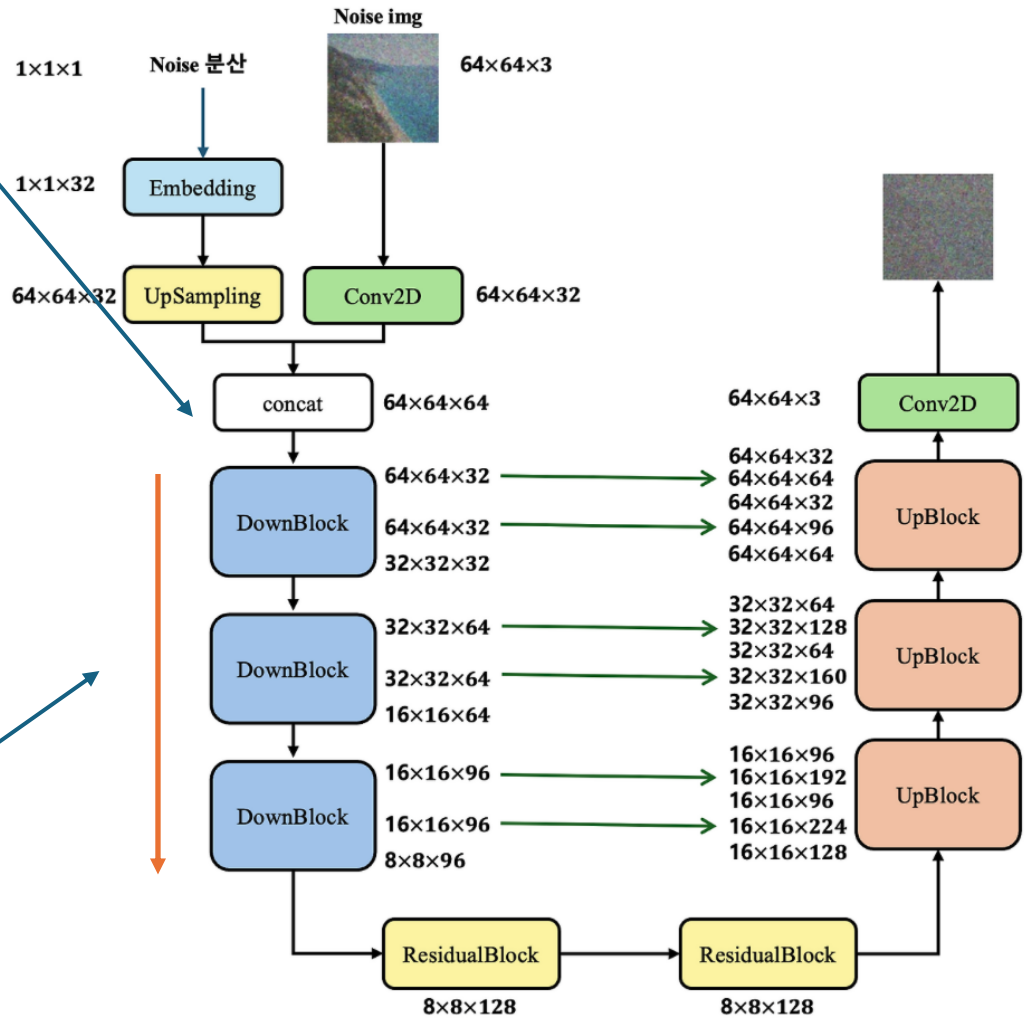
https://arxiv.org/pdf/2006.11239

Diffusion – DDPM Unet DownBlock

Timestep 정보 임베딩 + 노이즈 낀 이미지 처리



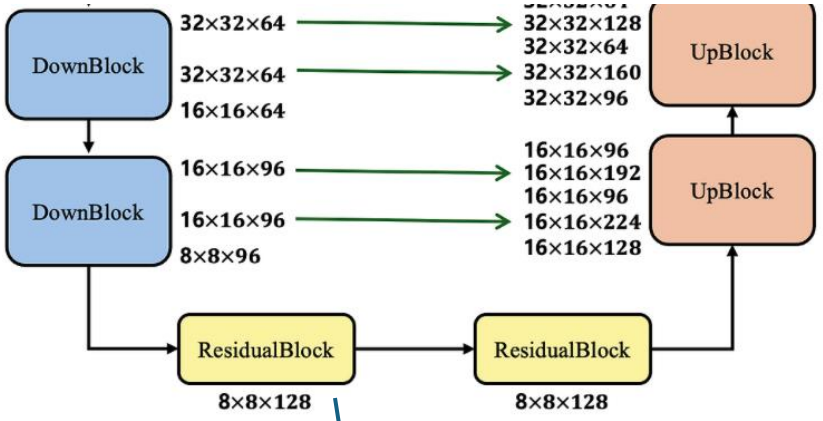
https://letter-night.tistory.com/207

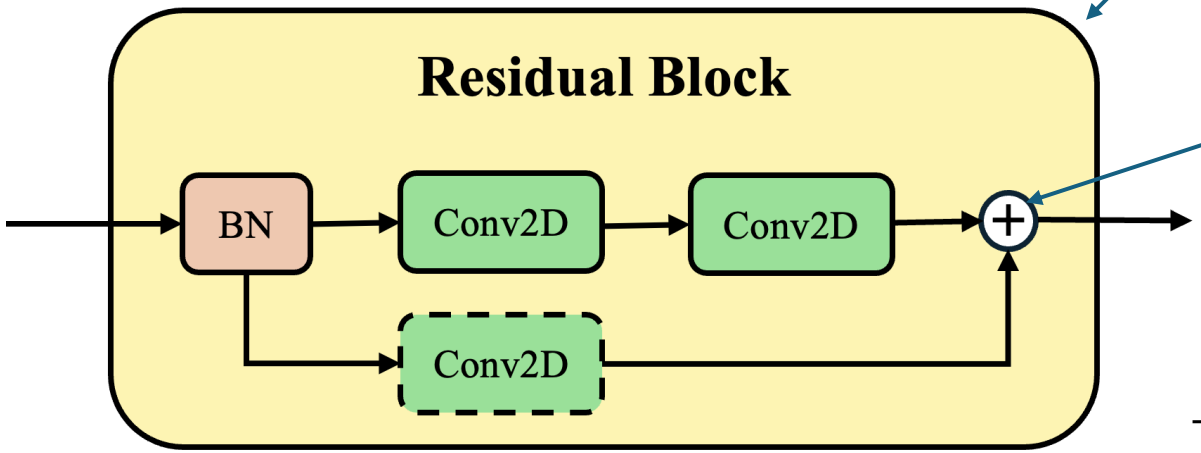


(Group) Normalization, self attention, activation, convolution 등 수행하며 특성 추출(채널 증가)

https://chan4im.tistory.com/271

Diffusion – DDPM Unet ResidualBlock





(단순화된 그림)

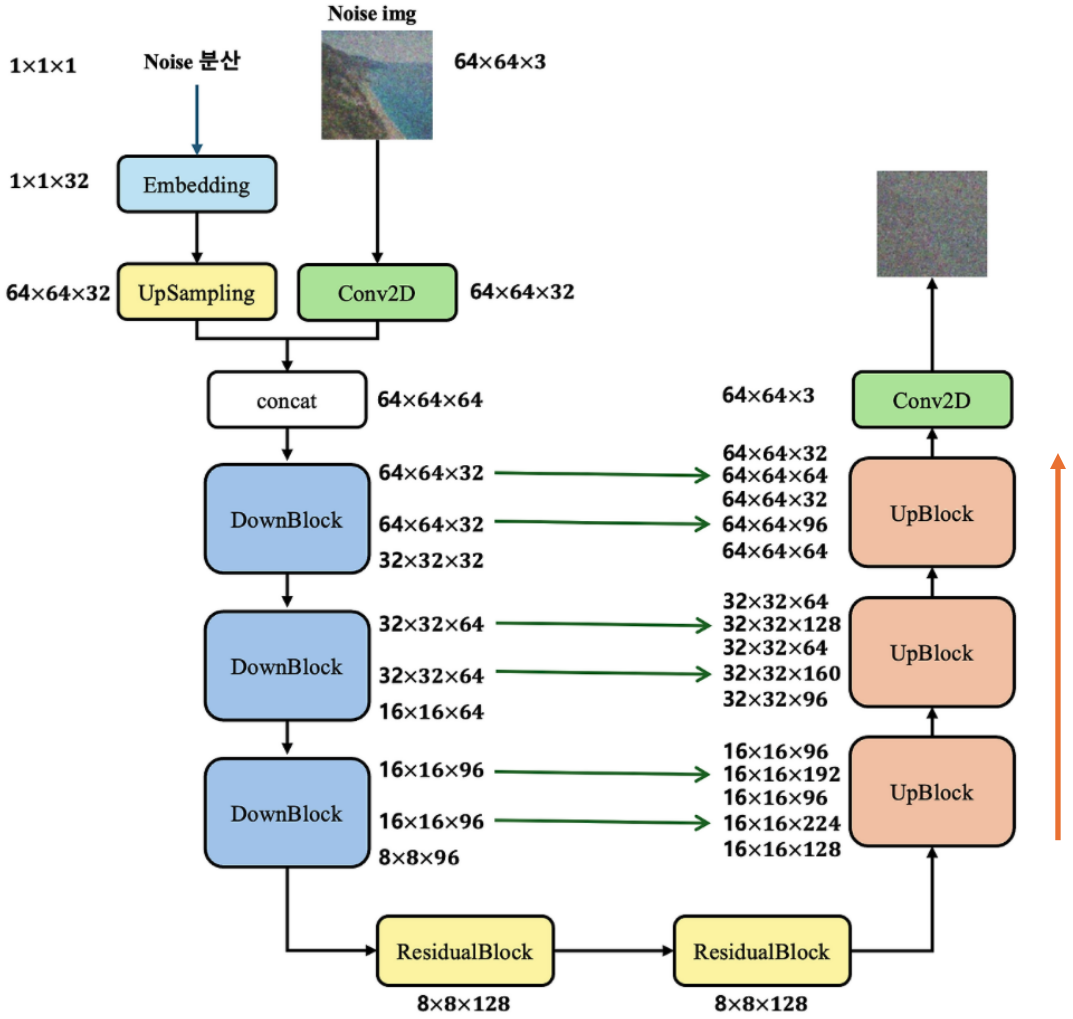
마찬가지로 convolution, normalization, activation등 진행.

Skip connection으로 정보 손실을 최소화

+ 그림과 다르게 실제로는 time embedding 값도 주입된다

https://chan4im.tistory.com/271

Diffusion – DDPM Unet UpBlock



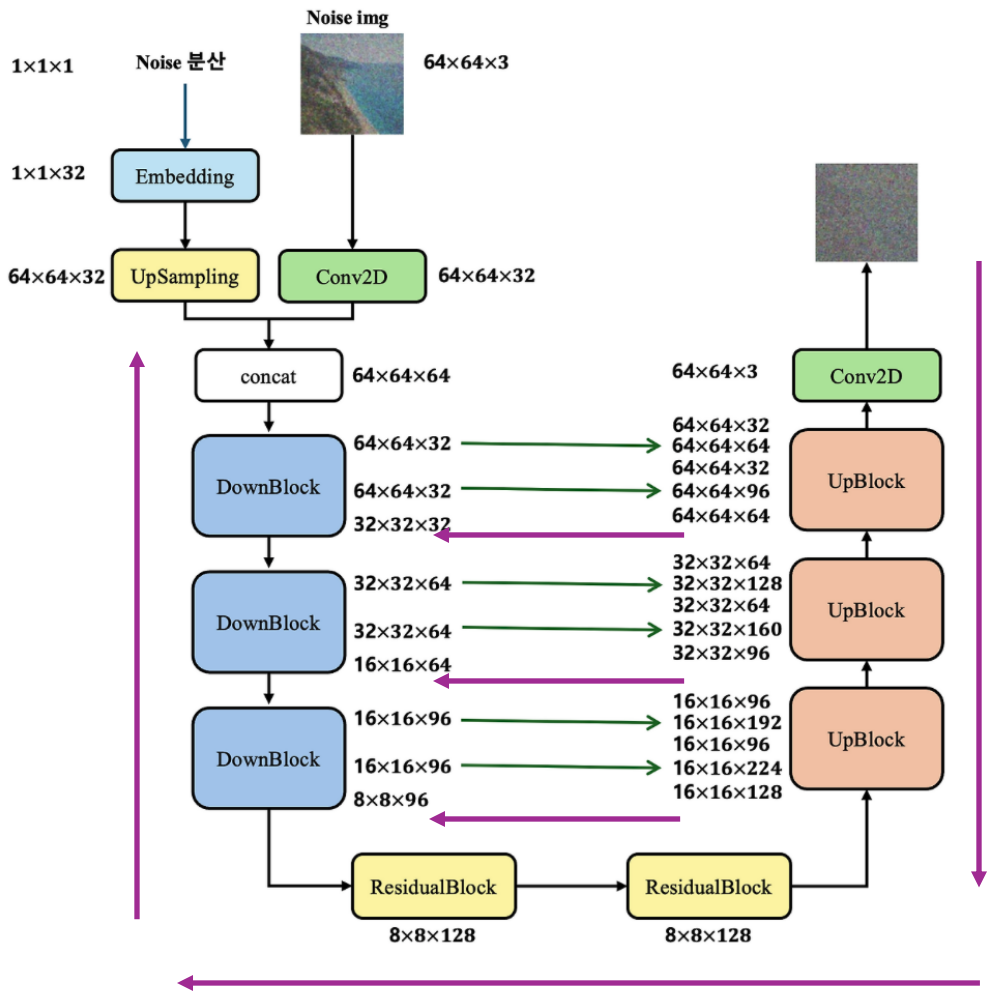
Downblock과 반대의 과정

채널을 줄이며 downblock에서
skip connected 된 정보들을 활용하여

ResidualBlock에서 받은 전역 정보
+ DownBlock의 local detail
살려서 noise 복원

https://chan4im.tistory.com/271

Diffusion – DDPM Unet BackProp

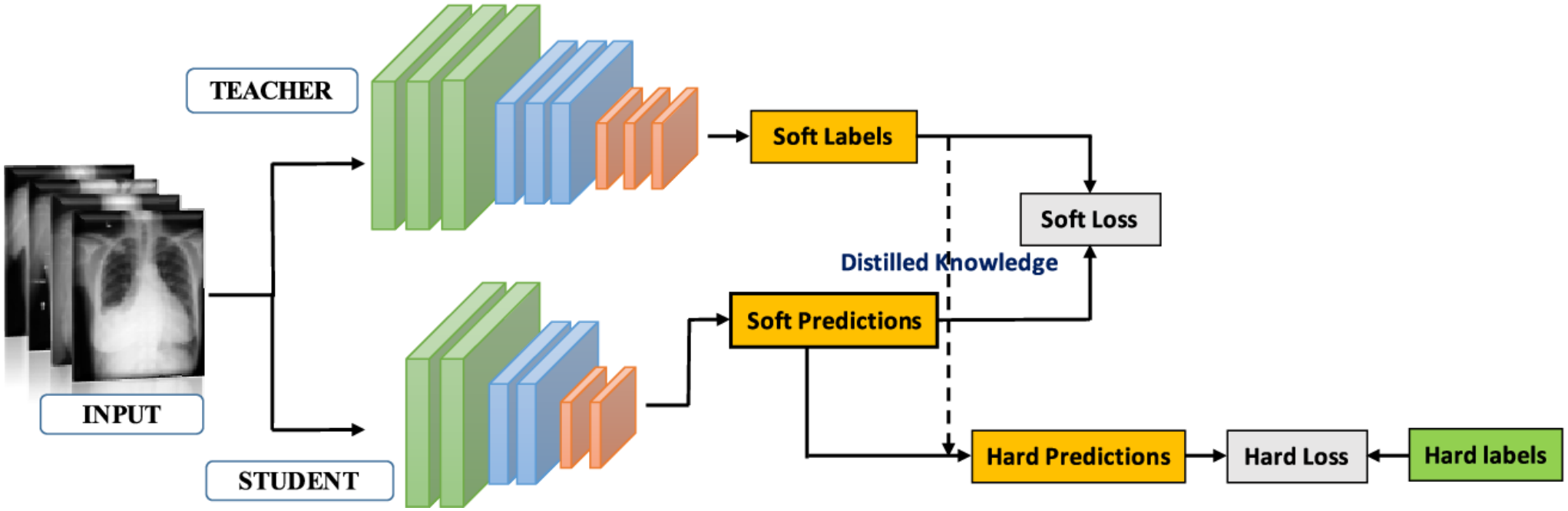


반대 방향으로 역전파 수행.
-> Update theta

Skip connection으로도 역전파가 전달된다.

https://chan4im.tistory.com/271

Distillation – What is Distillation?

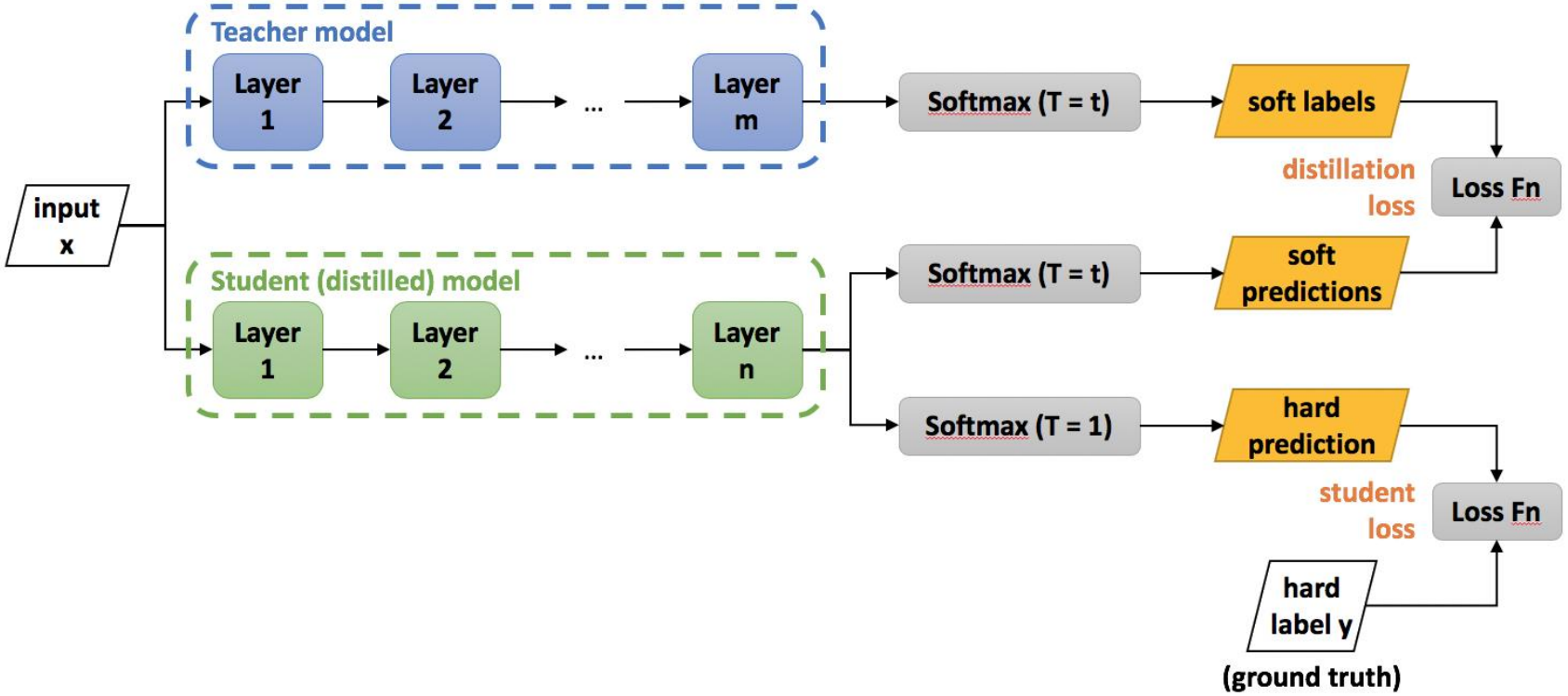


교사 모델(미리 학습된 모델)과 학생 모델(경량화 버전)준비

최종 목적: 학생 모델이 같은 input에 대해 교사 모델의 prediction을 학습

https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9183910

Distillation – Loss



교사 모델과는 soft prediction을 비교하여 distillation loss를 정의
Ground truth label과는 hard prediction을 비교하여 loss 정의

-> 적절히(하이퍼파라미터) 배합하여 하나의 식으로!

https://intellabs.github.io/distiller/knowledge_distillation.html

Distillation – Softmax Temperature

기본 softmax로
-> 잘 배합하여 하나의 식으로!

https://intellabs.github.io/distiller/knowledge_distillation.html

Pruning – What is Pruning?

Quantization – What is Quantization?